

Highlights

- Inverse-propensity and doubly robust graph propagation are biased beyond one hop.
- DRUP removes the repeated-walk bias exactly for any odd number of hops.
- Its guarantees need nuisances independent of the log, e.g. a sample split.
- Training-free DR propagation beats tuned trained models on KuaiRec.
- Thinning audits confirm exposure invariance on Coat and KuaiRec only.